# SMART INDIA HACKATHON 2026

## TITLE PAGE



- **Problem Statement ID:** SIH26204
- **Problem Statement Title:** Student Innovation - A solution/idea that can boost the current situation of the tourism industries including hotels, travel and others.
- **Theme:** Travel & Tourism
- **PS Category:** Software
- **Team ID:** Rawauthentic-SIH26
- **Team Name:** Rawauthentic
- **Proposed Solution:** SmartTour Connect (TourSetu AI) — An AI-Powered Smart Tourism, Direct Stay & Sustainable Travel Grid Platform

### SmartTour Connect

**NEXT-GEN AI TOURISM & TRANSIT GRID**

✓ 0% Commission Direct Hotel & Stay Bookings
✓ Dynamic AI Itinerary & Smart Budget Planner
✓ Real-time Crowd Density & Capacity Grid
✓ Instant QR Digital Boarding & Eco-Pass
✓ 24/7 Web Audio Hardware Emergency SOS
✓ Multi-Format Triple-Sync Ledger DB

# IDEA TITLE

## Proposed Solution :-

• SmartTour Connect is an AI-powered integrated digital platform designed to improve the tourism industry by connecting tourists, hotels, travel agencies, local guides, restaurants, transport providers, and local businesses on one smart platform.

• Improves tourist convenience, promotes local businesses and employment, reduces overcrowding, supports sustainable tourism, and creates a connected tourism ecosystem.

• AI Trip Planner: Personalized itineraries, hotels, transport, activities, and dynamic budgets.

• Tourism Hub: Connects tourists with verified hotels, guides, restaurants, and local businesses.

• Smart & Safe Travel: Provides crowd management, safety support, reviews, and smart recommendations.

## Problem Resolution :-

SmartTour Connect is an AI-powered one-stop platform for smarter travel planning. It provides personalized itineraries, hotel/activity recommendations, and budget planning. It connects tourists with verified local services and provides safety and crowd updates. It also promotes local businesses and enhances the overall travel experience.

## SmartTour Connect — Stakeholder Ecosystem & Modules

📱 **Traveler Dashboard & PWA:** Personalized recommendations, live weather, transit heatmaps, offline booking slips.

🤖 **AI Trip & Budget Planner:** Day-wise smart itineraries, expense breakdown, multi-currency conversion (₹, $, €, £, ¥).

🏨 **Direct Hotel & Stay Engine:** 0% OTA commission, direct host connection, dynamic pricing & room reservation.

🗺️ **Local Guides & Experiences:** Verified local experts, eco-trails, cultural food walks & artisan craft promotion.

🚨 **Smart Safety & Crowd SOS:** Real-time crowd heatmaps, hardware Web Audio SOS synthesizer & GPS distress beacon.

📊 **Admin & B2G Tourism Hub:** Central business verification, booking ledger management, state tourism analytics.

# TECHNICAL APPROACH

### Frontend (PWA)

- HTML5 & Tailwind CSS
- JavaScript (ES6+)
- Three.js (3D Earth)
- Leaflet.js (GIS Map)
- Web Audio (SOS Siren)

### Backend Server

- Python 3 Native HTTP
- Zero External Pip Libs
- REST API & CORS
- Threading Daemons
- Cloud Relay Worker

### Multi-Ledger DB

- backend_db.json (Store)
- backend_bookings.csv
- backend_summary.txt
- Triple-Sync Engine
- Zero Data Loss

### Edge Networking

- Cloudflare Tunnel
- HTTPS SSL/TLS
- Geolocation GPS API
- PWA Service Worker
- Git & GitHub CI/CD

## End-to-End Development & Implementation Flow

1. **Plan & Design** → Requirement analysis, user persona & Figma UI/UX architecture
2. **PWA Frontend** → Responsive UI, Three.js 3D Earth, Leaflet Maps, Web Audio SOS
3. **Python Backend** → Python 3 REST API, zero-dependency multithreaded server
4. **Triple-Sync DB** → Simultaneous save to JSON store, Excel CSV & Text receipts
5. **Security & Tunnel** → Cloudflare Zero-Trust edge tunneling for global HTTPS access
6. **Testing & CI/CD** → Cross-browser mobile PWA testing, GPS verification & GitHub versioning
7. **Deployment** → 1-click local launcher + edge cloud live deployment

# FEASIBILITY AND VIABILITY

## Feasibility :-

• Technical: Built on proven open-source AI/ML models, open APIs (OpenStreetMap, UPI), and auto-scaling cloud microservices → Highly scalable.

• Economic: Low-cost open-source tech stack; sustainable via micro-subscription for hotels and 0% commission direct bookings (saving 20-30% OTA cut).

• Operational: Dual light-interface: Zero-training PWA/portal for hoteliers & multi-lingual offline PWA mode for tourists.

## Potential Challenges & Mitigations :-

• **Spotty connectivity in remote tourist hubs**

→ Offline-first PWA with local edge caching for tickets & itineraries.

• **Tech hesitation among small homestay hosts**

→ Zero-friction 1-click WhatsApp & simple mobile onboarding.

• **Real-time crowd density accuracy**

→ GPS telemetry combined with historical algorithmic prediction.

## Viability :-

• Technically Feasible: Uses standard Python, WebGL & PWA APIs without requiring proprietary hardware.

• Economically Sustainable: Highly profitable for SMB hotels by replacing steep 20–30% OTA cuts with direct bookings.

• Socially Impactful: Decentralizes tourism revenue into rural & tribal belts while preventing degradation at fragile sites.

• Scalable & Adaptable: Plug-and-play architecture adaptable for State Tourism Boards, pilgrimage circuits, and heritage hubs.

### SUPPORTING FACTS FOR FEASIBILITY & VIABILITY

• Revenue Boost for Hotels: Hospitality analytics prove independent hotels using automated AI dynamic pricing see an 18%–25% increase in RevPAR and 20% higher off-season occupancy.

• Consumer Shift to Offbeat: UNWTO and Ministry of Tourism surveys indicate over 65% of Gen-Z & Millennial travelers actively seek offbeat local stays over crowded commercial hubs.

**Business Potential: Tiered SaaS for Hotels | Direct Booking Marketplace | B2G Analytics**

# IMPACT AND BENEFITS

## Potential impact on the target audience :-

- **Improves Tourist Experience:** Makes travel personalized, convenient, budget-friendly, and protected with offline emergency SOS.
- **Boosts Local Economy:** Increases visibility, direct bookings, and income for small homestays, rural hosts, and artisan guides.
- **Promotes Sustainable Tourism:** Reduces overcrowding via heatmaps and encourages exploration of lesser-known offbeat destinations.
- **Improves Decision-Making:** Provides actionable crowd and occupancy analytics for state tourism boards and business owners.

## Benefits of the Solution :-

- One-Stop Platform: Unifies itinerary planning, direct booking, multi-currency budgeting, and transit maps into a single PWA.
- AI-Powered Recommendations: Context-aware recommendations tailored to traveler budget, duration, and personal interests.
- Smart Budgeting: Live mathematical currency converter and expense categorization to eliminate hidden travel costs.
- Trusted & Safe Services: Verified host listings, contactless QR ticketing, and hardware emergency sound synthesizer.
- Promotes Local Tourism: Connects eco-tourists with indigenous homestays and authentic regional heritage.

## Unique Outcomes from Our Solution

**“SmartTour Connect transforms fragmented tourism services into one intelligent ecosystem — making travel personalized, affordable, safe, sustainable, and beneficial for both tourists and local businesses.”**

# RESEARCH AND REFERENCES

## • Research Papers & Citations:

**1. A Survey on Personalized Itinerary Recommendation**
Applied Soft Computing, 2024 | doi.org/10.1016/j.asoc.2023.111200

**2. How Smart is E-Tourism? Recommendation Systems**
Computer Science Review, 2021 | doi.org/10.1016/j.cosrev.2020.100337

**3. Artificial Intelligence in Tourism Research Agenda**
Sustainability, 2025 | doi.org/10.3390/su17209080

**4. Context-Aware Tourism using RAG & Semantic Re-Rank**
Electronics, 2025 | doi.org/10.3390/electronics14224448

## • Policy & Government References:

• ONDC Travel & Mobility Protocols (Open discovery network)
• Ministry of Tourism — India Tourism Statistics & Reports
• NITI Aayog — Rethinking Homestays & Rural Tourism Policy

## Platform Feature Comparison Matrix

| Feature / Capability | SmartTour | MakeMyTrip | ixigo | Wanderlog |
|---|---|---|---|---|
| 1. AI-Driven Dynamic Itinerary | ✓ (Full) | ✓ (Basic) | ✓ (Basic) | ✓ |
| 2. 0% OTA Host Commission | ✓ (Direct) | ✗ (20-30%) | ✗ (15-25%) | ✗ |
| 3. Local Homestay & Guide Hub | ✓ (Verified) | ✓ (Commercial) | ✓ | ✗ |
| 4. Real-Time Crowd Density Map | ✓ (Live) | ✗ | ✗ | ✓ (Limited) |
| 5. Hardware Emergency SOS Siren | ✓ (Offline) | ✗ | ✗ | ✗ |
| 6. Triple-Sync Multi-Format DB | ✓ (JSON/CSV) | ✗ | ✗ | ✗ |
| 7. Contactless QR Eco-Pass Pass | ✓ (Instant) | ✓ | ✓ | ✗ |

• **Research Flow: Problem ID → Literature Review → Gap Analysis → Tech Exploration → Validation → Proposed Solution → SmartTour Ecosystem**

• **Project Demo Links:** Web Portal: http://localhost:8000 | Admin Console: http://localhost:8000/admin.html | Presentation Visualizer: http://localhost:8000/sih_presentation.html